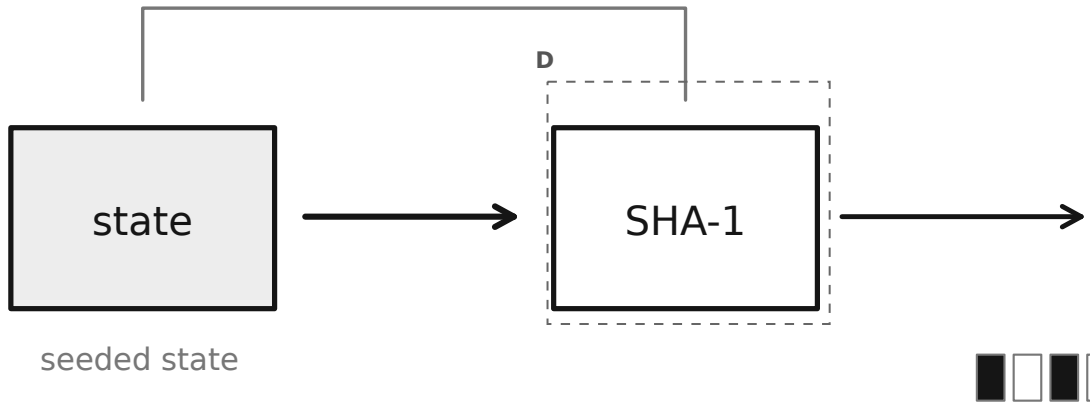
M166

Java SUN SHA1PRNG low-entropy seed replay

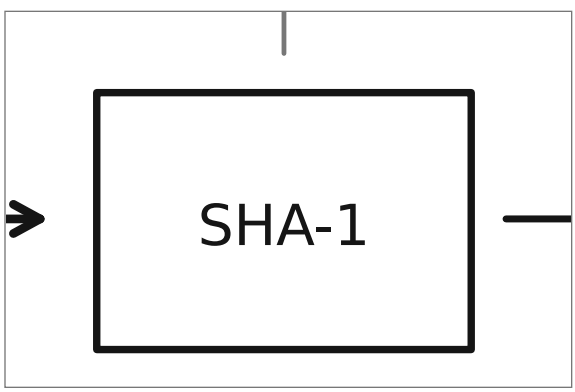


01 low-entropy seed

02 SUN SHA1PRNG state

03 deterministic bytes

D / OPERATION DETAIL



Vector detail / 1.42× main-view scale

HOW THE OPERATION WORKS

SUN SHA1PRNG with an explicit low-entropy seed evolves deterministic SHA-1-derived state and emits predictable bytes.

READ THE THREE STAGES

01 low-entropy seed
02 SUN SHA1PRNG state
03 deterministic bytes

WHAT WAS ACTUALLY DONE

The implementation reproduces setSeed byte order and signed state updates. It does not assume every SecureRandom call is weak; normal unobserved platform entropy changes the problem.

ORIGINAL FINDING / IMPLEMENTED/REGISTRY

Deterministic setSeed model is present; not every Java/Android wallet follows that path.